

DPEN022: Enabling Mathematics B

Class Questions Week 6A

Week 6A

Key Idea – Simple Substitution with other Functions

1. Use substitution to integrate the following:

(a) $\int \cos x \sqrt{\sin x - 1} \, dx$

(b) $\int \frac{e^{2x}}{(e^{2x} - 3)^7} \, dx$

(c) $\int e^{\cos x} \sin x \, dx$

(d) $\int x e^{\sin x^2} \cos x \, dx$

(e) $\int \frac{\sin x}{2 + \cos x} \, dx$

(f) $\int \frac{\sec^2(3x)}{1 + \tan(3x)} \, dx$

(g) $\int \frac{\cos x - \sin x}{\sin x + \cos x} \, dx$

(h) $\int \frac{e^{2x} + \cos(2x)}{e^{2x} + \sin(2x)} \, dx$

(i) $\int 2 \sin x \cos^6 x \, dx$

(j) $\int 3 \tan^{10} x \sec^2 x \, dx$

$$(a) \int \cos x \sqrt{\sin x - 1} \, dx$$

$$\text{Step 1: Let } u = \sin x - 1$$

$$\text{Step 2: } \frac{du}{dx} = \cos x \text{ and } du = \cos x \, dx$$

$$\begin{aligned} \text{Step 3: } \int \cos x \sqrt{\sin x - 1} \, dx &= \int \sqrt{\sin x - 1} \cos x \, dx \\ &= \int \sqrt{u} \, du \\ &= \int u^{\frac{1}{2}} \, du \end{aligned}$$

$$\text{Step 4: } = \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$= \frac{2u^{\frac{3}{2}}}{3} + C$$

$$\begin{aligned} \text{Step 5: } &= \frac{2(\sin x - 1)^{\frac{3}{2}}}{3} + C \\ &= \frac{2\sqrt{(\sin x - 1)^3}}{3} + C \end{aligned}$$

$$(b) \int \frac{e^{2x}}{(e^{2x} - 3)^7} \, dx$$

$$\int \frac{e^{2x}}{(e^{2x} - 3)^7} \, dx$$

$$\text{Step 1: Let } u = e^{2x} - 3$$

$$\text{Step 2: } \frac{du}{dx} = 2e^{2x} \text{ and } du = 2e^{2x} \, dx$$

$$\begin{aligned} \text{Step 3: } \int \frac{e^{2x}}{(e^{2x} - 3)^7} \, dx &= \frac{1}{2} \int \frac{1}{(e^{2x} - 3)^7} \times 2e^{2x} \, dx \\ &= \frac{1}{2} \int \frac{1}{u^7} \, du \\ &= \frac{1}{2} \int u^{-7} \, du \end{aligned}$$

$$\text{Step 4: } = \frac{1}{2} \times \frac{u^{-6}}{-6} + C$$

$$= -\frac{u^{-6}}{12} + C$$

$$\begin{aligned} \text{Step 5: } &= -\frac{(e^{2x} - 3)^{-6}}{12} + C \\ &= -\frac{1}{12(e^{2x} - 3)^6} + C \end{aligned}$$

(c) missing

(d) missing

$$(e) \int \frac{\sin x}{2 + \cos x} \, dx$$

$$\int \frac{\sin x}{2 + \cos x} \, dx \quad f(x) = 2 + \cos x$$

$$f'(x) = -\sin x$$

$$\begin{aligned} \int \frac{\sin x}{2 + \cos x} \, dx &= -\int \frac{-\sin x}{2 + \cos x} \, dx \\ &= -\ln|2 + \cos x| + C \end{aligned}$$

$$(f) \int \frac{\sec^2(3x)}{1 + \tan(3x)} \, dx$$

$$\int \frac{\sec^2(3x)}{1 + \tan(3x)} \, dx \quad f(x) = 1 + \tan(3x)$$

$$f'(x) = 3\sec^2(3x)$$

$$\begin{aligned} \int \frac{\sec^2(3x)}{1 + \tan(3x)} \, dx &= \frac{1}{3} \int \frac{3\sec^2(3x)}{1 + \tan(3x)} \, dx \\ &= \frac{1}{3} \ln|1 + \tan(3x)| + C \end{aligned}$$

$$(g) \int \frac{\cos x - \sin x}{\sin x + \cos x} dx$$

$$\int \frac{\cos x - \sin x}{\sin x + \cos x} dx \quad f(x) = \sin x + \cos x$$

$$f'(x) = \cos x - \sin x$$

$$\int \frac{\cos x - \sin x}{\sin x + \cos x} dx = \ln |\sin x + \cos x| + C$$

$$(h) \int \frac{e^{2x} + \cos(2x)}{e^{2x} + \sin(2x)} dx$$

$$\int \frac{e^{2x} + \cos(2x)}{e^{2x} + \sin(2x)} dx \quad f(x) = e^{2x} + \sin(2x)$$

$$f'(x) = 2e^{2x} + 2\cos(2x)$$

$$f'(x) = 2(e^{2x} + \cos(2x))$$

$$\int \frac{e^{2x} + \cos(2x)}{e^{2x} + \sin(2x)} dx = \frac{1}{2} \int \frac{2(e^{2x} + \cos(2x))}{e^{2x} + \sin(2x)} dx$$

$$= \frac{1}{2} \ln |e^{2x} + \sin(2x)| + C$$

$$(i) \int 2 \sin x \cos^6 x \, dx = 2 \int \sin x (\cos x)^6 \, dx$$

$$\int 2 \sin x \cos^6 x \, dx = 2 \int \sin x (\cos x)^6 \, dx$$

Step 1: Let $u = \cos x$

$$\text{Step 2: } \frac{du}{dx} = -\sin x \text{ and } du = -\sin x \, dx$$

$$\text{Step 3: } 2 \int \sin x (\cos x)^6 \, dx = -2 \int (\cos x)^6 (-\sin x) \, dx \\ = -2 \int u^6 \, du$$

$$\text{Step 4: } = -2 \times \frac{u^7}{7} + C$$

$$= -\frac{2u^7}{7} + C$$

$$\text{Step 5: } = -\frac{2(\cos x)^7}{7} + C$$

$$(j) \int 3 \tan^{10} x \sec^2 x \, dx$$

$$\int 3 \tan^{10} x \sec^2 x \, dx = \int 3 (\tan x)^{10} \sec^2 x \, dx$$

Step 1: Let $u = \tan x$

$$\text{Step 2: } \frac{du}{dx} = \sec^2 x \text{ and } du = \sec^2 x \, dx$$

$$\text{Step 3: } \int 3 (\tan x)^{10} \sec^2 x \, dx = \int 3u^{10} \, du$$

$$\text{Step 4: } = \frac{3u^{11}}{11} + c$$

$$\text{Step 5: } = \frac{3(\tan x)^{11}}{11} + c$$

$$\int_0^{\ln 2} e^x \sqrt{e^x + 1} \, dx$$

Step 1: Let $u = e^x + 1$

Limits: For $x = \ln 2$, $u = e^{\ln 2} + 1 = 2 + 1 = 3$

For $x = 0$, $u = e^0 + 1 = 1 + 1 = 2$

Step 2: $\frac{du}{dx} = e^x$ and $du = e^x dx$

Step 3: $\int_0^{\ln 2} e^x \sqrt{e^x + 1} \, dx = \int_2^3 (e^x + 1)^{\frac{1}{2}} e^x dx$

$$= \int_2^3 u^{\frac{1}{2}} du$$

Step 4:

$$= \left[\frac{u^{\frac{3}{2}}}{\frac{3}{2}} \right]_2^3$$

$$= \left[\frac{2u^{\frac{3}{2}}}{3} \right]_2^3$$

$$= \frac{2 \times 3^{\frac{3}{2}}}{3} - \frac{2 \times 2^{\frac{3}{2}}}{3}$$

$$= \frac{2\sqrt{3^3} - 2\sqrt{2^3}}{3}$$

$$= \frac{2\sqrt{3^2} \sqrt{3} - 2\sqrt{2^2} \sqrt{2}}{3}$$

$$= \frac{6\sqrt{3} - 4\sqrt{2}}{3}$$

$$\int_0^{\frac{\pi}{4}} 3 \tan^{10} x \sec^2 x \, dx = \int_0^{\frac{\pi}{4}} 3 (\tan x)^{10} \sec^2 x \, dx$$

Step 1: Let $u = \tan x$

Limits: For $x = \frac{\pi}{4}$, $u = \tan \frac{\pi}{4} = 1$

For $x = 0$, $u = \tan 0 = 0$

Step 2: $\frac{du}{dx} = \sec^2 x$ and $du = \sec^2 x \, dx$

Step 3: $\int_0^{\frac{\pi}{4}} 3 (\tan x)^{10} \sec^2 x \, dx = \int_0^1 3u^{10} \, du$

Step 4:

$$= \left[\frac{3u^{11}}{11} \right]_0^1$$

$$= \frac{3 \times 1^{11}}{11} - \frac{3 \times 0^{11}}{11}$$

$$= \frac{3}{11}$$

$$\int (2x+1)e^{x^2+x+1} dx$$

Step 1: Let $u = x^2 + x + 1$

Step 2: $\frac{du}{dx} = 2x + 1$ and $du = (2x + 1) dx$

Step 3: $\int (2x+1)e^{x^2+x+1} dx = \int e^{x^2+x+1} (2x+1) dx$
 $= \int e^u du$

Step 4: $= e^u + C$

Step 5: $= e^{x^2+x+1} + C$